

Real Time Oral Cavity Detection Leading to Oral Cancer using CNN

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Abstract— Oral cancer presents a significant burden to global health, characterized by rising incidence rates and the associated mortality it entails. Timely identification and intervention hold immense potential in enhancing patient prognosis in the face of this disease. Nevertheless, conventional diagnostic approaches often hinge on subjective visual assessments by healthcare professionals, which can lead to potential delays in detection and diagnosis. The realm of medical image analysis has witnessed a transformative shift in recent times with the advent of deep learning methodologies. By harnessing the capabilities of convolutional neural networks (CNNs), scientists have made remarkable strides in employing these models to address diverse medical imaging tasks, including the early detection of cancerous conditions across different anatomical locations. In this paper, we propose a groundbreaking approach for the real-time identification of oral cavity conditions, with a specific emphasis on predicting the occurrence of oral cancer through the utilization of a deep learning framework. Our methodology synergistically integrates patient questionnaires and oral cavity images, culminating in a prediction model that enhances accuracy and reliability. Through this user-friendly system, individuals can undergo oral cancer diagnosis without invasive techniques, making early detection more accessible and potentially life-saving. The integration of communication with healthcare professionals further enhances the overall process, enabling users to seek timely guidance and consultation for better management of their oral health. This streamlined approach empowers individuals to actively participate in their healthcare, bridging the gap between medical technology and patient-centric care.

Keywords - Convolutional Neural Networks, Deep Learning, Oral Cancer, Oral Potentially Malignant Disorder.

I. INTRODUCTION

Oral cancer presents a significant burden to global health, characterized by rising incidence rates and the associated mortality it entails. Timely identification and intervention hold immense potential in enhancing patient prognosis in the face of this disease. Nevertheless, conventional diagnostic approaches often hinge on subjective visual assessments by healthcare professionals, which can lead to potential delays in detection and diagnosis. The realm of medical image analysis has witnessed a transformative shift in recent times with the advent of deep learning methodologies. By harnessing the capabilities of convolutional neural networks (CNNs), scientists have made remarkable strides in employing these models to address diverse medical imaging tasks, including

the early detection of cancerous conditions across different anatomical locations. Inspired by the intricate interconnections of neurons in the human brain, a computational framework gives birth to the concept of neural networks. These networks comprise multiple layers of artificial nodes or units, meticulously processing and transmitting information. Through a rigorous training regimen, they acquire the ability to discern patterns and unravel intricate relationships in data, thereby empowering them to make accurate predictions and classifications. The transformative potential of neural networks has been harnessed across diverse domains, including image recognition, natural language processing, and

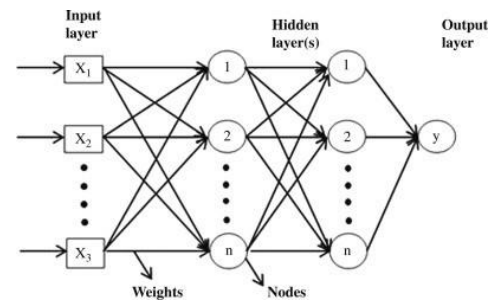


Fig. 1. Neural network architecture

the realm of autonomous driving. Convolutional Neural Networks (CNNs) have emerged as a valuable tool for analyzing medical images, facilitating the diagnosis of oral cancer. A typical neural network architecture is shown in Fig.1 [1]. By training a CNN on an extensive dataset of oral cancer images, it becomes adept at identifying cancerous regions with remarkable precision. This breakthrough empowers healthcare professionals with early detection capabilities, aids in treatment planning, and provides a means for monitoring oral cancer patients effectively.

In the present study, we introduce a groundbreaking approach for the real-time identification of oral cavity conditions, with a specific emphasis on predicting the occurrence of oral cancer through the utilization of a deep learning framework. Our methodology synergistically integrates patient questionnaires and oral cavity images, culminating in a prediction model that enhances accuracy and reliability.

A. Organization of Paper

The rest of the paper is organized as Section 2 delves into the pertinent literature review, Section 3 presents our innovative

system proposal, Section 4 outlines the materials and methods employed, Section 5 elucidates the results and subsequent discussion, Section 6 provides visual representations of the user interface through snapshots, and finally, Section 7 encapsulates the conclusive remarks of this paper.

II. RELATED WORK

As per research findings in reference [2], the implementation of advanced deep learning methods proves instrumental in enabling timely identification and proactive measures against oral cancer. These cutting-edge techniques empower healthcare professionals to accurately interpret crucial medical imaging data, enhancing the efficiency of oral cancer detection. This transformative approach holds promise for cost-effective and clinically relevant interventions, ensuring non-invasive identification of oral cancer at an early stage, thereby emphasizing its significance. In this particular research, we employed a distinctive Convolutional Neural Network (CNN) architecture to effectively classify intraoral images. Our carefully curated dataset consisted of 598 images, which we divided into two subsets: 402 for training purposes and 196 for testing. Through our sequential model, we successfully categorized these images into three distinct classes: 'Normal teeth', 'Caries', and 'Oral cancer'. Our disease detection accuracy on tested tooth photos demonstrated an impressive range of 83 percent to 94 percent. The system, as described in [3], plays a crucial role in enabling early diagnosis of oral concerns, providing dentists with efficient and accurate identification of dental health problems without requiring direct examination. In another enlightening paper [4], we propose an innovative approach utilizing an ensemble of pre-trained deep-learning models for the early detection of OCSCC (oral cavity squamous cell carcinoma) from microscopic images. The methodology incorporates shared layers to unify the ensemble heads, and we leverage transfer learning by utilizing various pre-trained models with the assistance of the Augmentor library to generate high-quality datasets. To validate the approach, we compare our results with those of another article that employed the same dataset, thereby affirming the potential and promise of our proposed methodology for OCSCC detection and classification. The primary focus of the compelling paper [5] is to introduce a novel methodology that harnesses the power of deep learning techniques for the early detection of skin and oral cancer. Within the paper, readers will find an in-depth literature review that encompasses existing research papers relating to the detection of these cancers. Additionally, we provide a visually informative flowchart that illustrates the proposed methodology, enhancing comprehension for readers. Within the research paper [6], our attention is centered on the comprehensive analysis and prediction of oral cancer using a diverse range of machine learning classifiers. We implement these classifiers with meticulously selected datasets and conduct thorough evaluations of their performance. The results demonstrate the effectiveness of our approach in accurately classifying various stages of carcinoma. By leveraging machine learning methods, we enable early analysis and detection of carcinoma, and the amalgamation of different techniques exhibits promising potential for early diagnosis.

It is important to note that experienced clinicians should face no difficulty in identifying precursor lesions that typically precede oral cancer, as they often present with distinct clinical features. This paper also presents the 2017 World Health Organization (WHO) classification system for 'oral potentially malignant disorders', which encompasses a range of conditions associated with habits, immune-mediated or inflammatory factors, solar radiation-induced conditions such as actinic cheilitis, and genetic disorders like dyskeratosis congenital. In this paper [7], we delve into the exploration of common risk factors and oral potentially malignant disorders, shedding light on these crucial aspects.

In response to the urgent demand for automated detection of oral ailments, a groundbreaking solution emerged, encompassing the utilization of smartphone-derived imaging and cutting-edge deep learning algorithms [8]. This method employed a simple yet effective technique for capturing oral cavity images, emphasizing a centered rule for image capture. To combat the variability arising from smartphone cameras, a medium-sized dataset encompassing five disease categories was meticulously curated, and a resampling technique was introduced. Evaluating the efficiency of this approach in oral cancer detection, the state-of-the-art deep learning network known as HRNet was employed. The potential of automating early detection of both potentially malignant and malignant oral cavity lesions holds promise for cost-effective disease diagnosis. The MeMoSA project (Mobile Mouth Screening Anywhere) plays a pivotal role in building a comprehensive collection of meticulously annotated oral lesions by sourcing images from clinical experts worldwide and employing a dedicated annotation tool for precise labeling. An innovative approach presented in reference [9] combines bounding box annotations contributed by multiple clinicians. By harnessing deep neural networks, intricate patterns can be extracted, effectively addressing the intricacies of this task. The study demonstrated encouraging results by employing two deep learning-based computer vision approaches—ResNet-101 for image classification and Faster R-CNN for object detection.

The main objective of this study is to develop an automated system for early detection of potentially malignant oral lesions. A crucial aspect of this work entails creating a comprehensive dataset of meticulously annotated oral lesions. In their research paper [10], the authors propose an innovative approach to construct automatic oral cancer image classification software. Leveraging the gathered data, machine learning models, and deep neural networks such as Naive Bayes, KNN, SVM, ANN, and a newly designed 43-layer CNN network inspired by VGG-16. To enhance treatment outcomes and prevent malignant transformation, early detection and accurate staging of oral cancer are of paramount importance. This paper provides a comprehensive overview of standardized protocols for collecting oral samples, characterizing them, and conducting molecular risk assessments. By offering guidance on suitable sampling methods and tailored downstream analyses aligned with specific research objectives, researchers can gain valuable insights from the information shared in this paper [11]-[16].

III. PROPOSED SYSTEM

A. Existing System

- Dentists and healthcare professionals use visual examination to carefully inspect the oral cavity, looking for any unusual changes in color, texture, or shape in the lips, gums, tongue, and other oral tissues to identify potential areas of concern.
- Non-invasive techniques like brush biopsy, toluidine blue staining, VELscope, and fluorescence imaging are employed to aid in the early detection of oral abnormalities and cancerous lesions. These methods help dentists visualize and evaluate suspicious areas, facilitating early intervention and appropriate treatment planning.

The workflow of the proposed system is shown in Fig.2 .

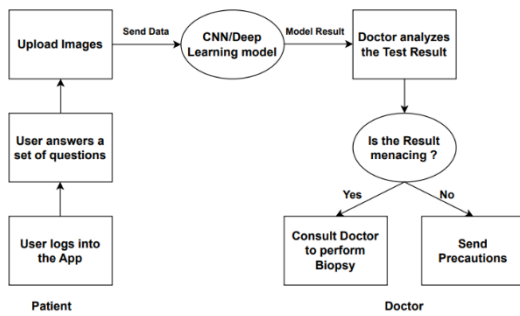


Fig. 2. System workflow

The mobile application, compatible with both Android and iOS platforms, empowers users with a range of functionalities, including:

- Seamless registration and login process.
- Access to a comprehensive oral health assessment test, comprising:
 1. Respond to a carefully curated questionnaire.
 2. Capturing clear and precise image(s) of the oral cavity.
- The accumulated data is then securely transmitted to the Django REST server. Utilizing cutting-edge deep learning algorithms, the server leverages a trained model to analyze the loaded oral cavity image and predict its corresponding class. To enhance accuracy, the model takes three images as input and calculates the average probability for each class. Additionally, the user's responses to the questionnaire are evaluated by mapping them to predetermined options for every question. This evaluation process involves calculating the probability for each class, providing further insights into the user's oral health status. The proposed system introduces more advanced and accurate methods for oral health assessment, leveraging technology and automation to enhance the detection and evaluation of potential oral health issues. It aims to provide users with a more convenient and reliable tool to monitor their oral health and seek timely professional assistance when needed.

B. Scope of the System

- The application will enable even a normal person (non-medical professional) to detect early changes

in the oral cavity using her/his device's camera and by answering a few questions about their daily habits.

- The application will help the oncologist/dentist to understand the severity of the lesion and localize the region for biopsy.

C. Materials and Methods

The depicted illustration in Fig.3 provides a comprehensive overview of the proposed undertaking.

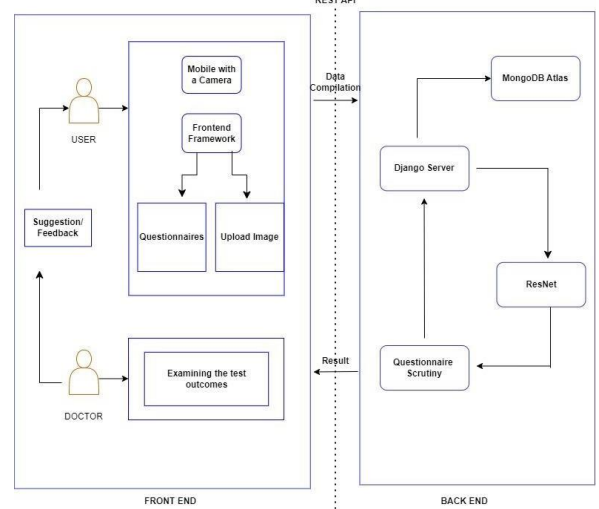


Fig. 3. High level design

Using a mobile application accessible on Android and iOS platforms, individuals are empowered to engage in an all-encompassing evaluation of their oral cavity well-being. This interactive process entails responding to inquiries and capturing images of the oral cavity for assessment purposes, while also facilitating user registration and login functionalities.

Data is sent to a Django REST server, where a deep learning model predicts the class of the images and calculates average probabilities. The user's questionnaire responses are mapped to probabilities, and the combined results are weighted towards the questionnaire. The final result is analyzed by the doctor and displayed in the mobile application for the user.

ResNet152 Model Initialization and Setup are as shown in the below algorithm:

- 1: **Input:** Pretrained model
- 2: **Output:** Trained model
3. $\text{model} \leftarrow \text{models.resnet152}(\text{pretrained} = \text{True})$
4. $\text{num_ftsr} \leftarrow \text{model.fc.in_features}$
5. $\text{model.fc} \leftarrow \text{nn.Linear}(\text{num_ftsr}, 2)$
6. $\text{model} \leftarrow \text{model.to}(\text{device})$
7. $\text{criterion} \leftarrow \text{nn.CrossEntropyLoss}()$
8. $\text{optimizer} \leftarrow \text{torch.optim.Adam}(\text{model.parameters}(), \text{lr} = 0.001)$

1) Oral Dataset

The training dataset used for the ResNet152 model includes two main components: data collected from the website and real-time datasets from a medical college. This combination ensures a diverse and comprehensive range of data for training the model.

The website data provides a broad representation of cases, while the real-time dataset from the medical college offers specific and up-to-date information. By incorporating both sources, the model can learn from a wide array of scenarios, enhancing its accuracy and robustness.

The inclusion of real-time medical data not only enriches the training process but also ensures that the model is trained on relevant and current information, making it closely aligned with practical application in the field of oral health and cancer detection.



Fig. 4. Cancer image



Fig. 5. Non-cancer image

The cancer images exhibit distinct lesions characterized by abnormal growths, ulcerations, or malignant formations within the oral cavity. These lesions serve as crucial indicators of potential malignancy as shown in Fig.4.

On the other hand, non-cancer images in this study primarily consist of ulcers, which are open sores or lesions caused by various factors but do not possess malignant characteristics as shown in Fig.5.

2) Training and Testing Methodology

A collection of oral cancer images is partitioned into two sets, namely training, and testing, following a conventional 80/20 division. The dataset consists of 150 images of both cancer and non-cancer. The training set plays a crucial role in training the CNN model, wherein the images are passed through the network, and the model's weights are fine-tuned to minimize prediction errors. After the completion of the training process, the testing set is employed to gauge the performance of the trained model. Various metrics, including accuracy and precision, are measured to assess the efficacy of the model.

D. Network Architecture

1) VGG19

The VGG19 model architecture consists of a total of 19 layers, encompassing 16 convolutional layers and 3 fully connected layers. The convolutional layers effectively utilize 3x3 filters with a stride of 1, while the max-pooling layers employ 2x2 filters with a stride of 2. This deep network's complexity is further enhanced by a progressive increment in the number of filters, initiating at 64 and doubling after each max-pooling layer.

2) GoogLeNet

GoogLeNet utilizes a deep neural network architecture with 22 layers. The key component is the inception module, which combines 1x1, 3x3, and 5x5 convolutional filters in parallel to capture features at different scales. It also employs average pooling and max-pooling layers, with additional auxiliary classifiers to encourage intermediate feature learning.

3) ResNet101

ResNet101 is a deep neural network structure comprising 101 layers, incorporating residual connections that enable the learning of residual mappings. The fundamental component is the residual block, employing two 3x3 convolutional layers with shortcut connections. The convolutional layers progressively increase the number of filters, beginning with 64 and doubling after each down-sampling step.

4) ResNet152

ResNet152 is an extension of ResNet101, with 152 layers as depicted in Fig.6. It follows the same residual block structure but with increased depth. It also starts with 64 filters and doubles the number of filters after each down-sampling layer. The additional layers in ResNet152 enable it to capture more intricate features and improve the representation capacity of the network.

Two applications of ResNet152 are:

- **Image Classification:** ResNet152 has been widely used in image classification tasks, such as the ImageNet Large Scale Visual Recognition Challenge (ILSVRC), where it has achieved top performance by accurately classifying images into various categories. Its deep architecture and skip connections enable it to capture intricate features and improve classification accuracy.
- **Object Detection:** ResNet152 has also been applied in object detection tasks, where the goal is to identify and localize multiple objects within an image. By utilizing the deep features learned by ResNet152, object detection models can leverage the network's representation capacity to detect objects with higher accuracy and precision, contributing to improved object detection performance.

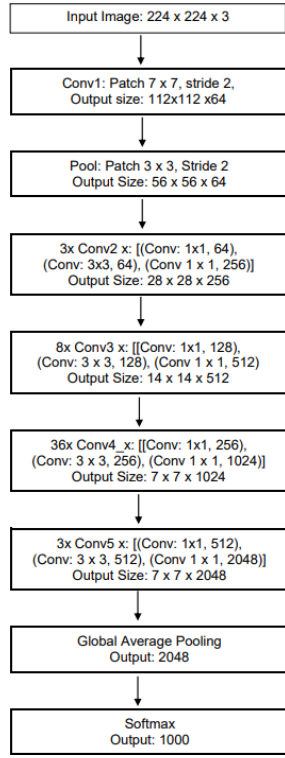


Fig. 4. ResNet152 architecture

Table 1 provides a comparison of four popular convolutional neural networks (CNN) models: VGG19, GoogLeNet, ResNet101 and ResNet152. Each model has its unique architecture and characteristics. The table includes details such as the number of parameters, number of layers, filter size, input size, and output size for each model. This information allows for a quick overview of the key specifications of these CNN architectures.

TABLE I. COMPARISON OF DIFFERENT CNN MODELS

Model	Layers	Accuracy	No. of parameters
VGG19	19	69.44%	143 million
GoogLeNet	22	63.89%	6.8 million
ResNet101	101	72.22%	44.6 million
ResNet152	152	80.56%	60.3 million

E. Image Processing and Classification

The workflow of the entire system is shown in Fig. 7.

- Image resizing and normalization are common image enhancement techniques used to standardize the input oral cavity images for further analysis, ensuring consistent size and pixel value ranges.
- VGG19, GoogLeNet, ResNet101, and ResNet152 are popular pre-trained deep learning models employed for feature extraction in transfer learning, leveraging their learned features from large-scale datasets to extract meaningful representations from oral cavity images.

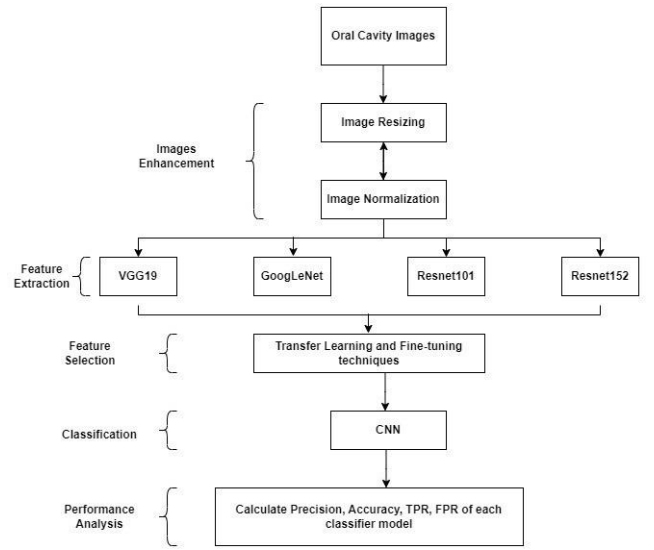


Fig. 5. Work flow

- Transfer learning and fine-tuning techniques exploit the pre-trained models' knowledge by adjusting and fine-tuning their parameters on the oral cavity image dataset, enabling efficient feature selection and improving the classification performance of the CNN models. Accuracy, precision, true positive rate (TPR), and false positive rate (FPR) are commonly evaluated performance metrics to assess the classification performance of these models.

IV. RESULTS AND DISCUSSION

A. System and Software Requirements

To meet the hardware requirements for mobile app development and testing, a computer system with a Quad-Core processor of 2.0 GHz or greater, at least 16 GB of RAM, and 20 GB of available hard disk space is needed. Additionally, input devices like a keyboard, mouse, and web camera are essential. An Android phone with an OS later than Jellybean is required for testing Android apps. For app development, a laptop or desktop computer with sufficient processing power and memory is necessary, along with a stable internet connection for downloads, online resources, and collaboration. Additional hardware like a USB cable and an external hard drive may be necessary. The software requirements include Windows 10 or newer, Django, Jupyter Notebook, Postman API, Tensorflow, NodeJS, Visual Studio Code, Flutter, Android Studio, and MongoDB Atlas for cloud-based database tasks. This comprehensive setup enables efficient and effective mobile app development and testing.

B. Experimental Evaluation

In this section, to select the optimal model for oral cancer detection, we considered three important metrics: confusion matrix, accuracy, and the ROC (Receiver Operating Characteristic) curve.

1) Confusion Matrix

The depicted representation in Fig.8, known as the confusion matrix, offers a holistic and detailed overview of the model's effectiveness. It provides a comprehensive account of the number of accurate positive predictions, accurate negative predictions, erroneous positive predictions, and erroneous negative predictions. This matrix serves as a crucial tool in assessing accuracy and precision, enabling a thorough evaluation of the predictive capabilities of each model. Consequently, meaningful performance comparisons can be made by analyzing the unique insights provided by the confusion matrix.

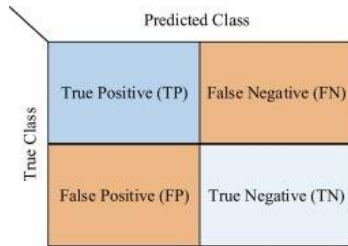


Fig. 6. Confusion Matrix

Observing Fig.9, we are presented with a bar graph that efficiently compares the false positive (FP) and false negative (FN) counts for four distinct models: VGG19, GoogLeNet, ResNet101, and ResNet152. The x-axis corresponds to the model names, while the y-axis quantifies the number of misclassifications. Each bar's height represents the respective count of false positives and false negatives about each model.

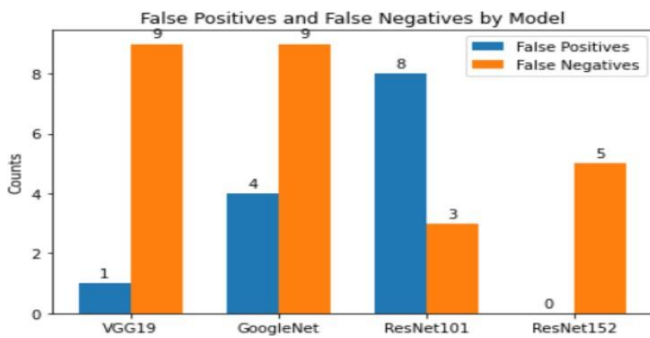


Fig. 9. Bar graph of the comparison of different models

Upon meticulous examination, an intriguing pattern unfolds. ResNet152 conspicuously boasts the fewest false positives and false negatives in comparison to the other models. This finding implies that ResNet152 exhibits superior accuracy in precisely categorizing instances of oral cancer, thus resulting in a reduced occurrence of misclassifications. Conversely, VGG19, GoogLeNet, and ResNet101 demonstrate varying degrees of misclassifications, indicated by higher counts of false positives and false negatives.

2) Accuracy

The assessment of accuracy is determined by dividing the count of precise predictions by the total number of predictions conducted. This measurement represents the proportion of accurately categorized instances from the entire dataset, as exemplified in equation 1.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (1)$$

3) ROC Curve

In Fig.10, the ROC curve illustrates the equilibrium between sensitivity (true positive rate) and specificity (1 - false positive rate) at various classification thresholds. The area under the ROC curve (AUC) encapsulates the model's discriminatory capability, with a greater AUC indicating superior overall performance. Through a comparison of ROC curves across different models, the optimal model with the highest AUC can be discerned.

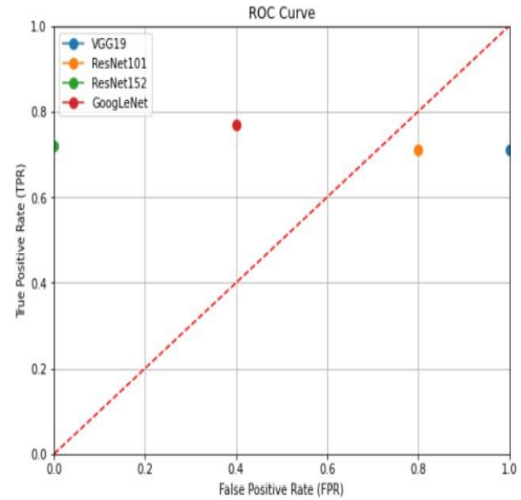


Fig. 10. ROC curve

The presented results showcase the ROC curve, depicted in Fig.10, elucidating the distinction between false positive rates along the x-axis and true positive rates along the y-axis. Each point on this graph corresponds to a distinct classification threshold, visually representing the models' ability to differentiate between positive and negative instances. A more elevated curve signifies superior performance. Notably, upon careful analysis, it becomes apparent that ResNet152 surpasses the other models by exhibiting a higher true positive rate at lower false positive rates. This implies that ResNet152 achieves a commendable equilibrium between sensitivity and specificity in the detection of oral cancer.

In Fig.11, the accompanying bar chart demonstrates the Area

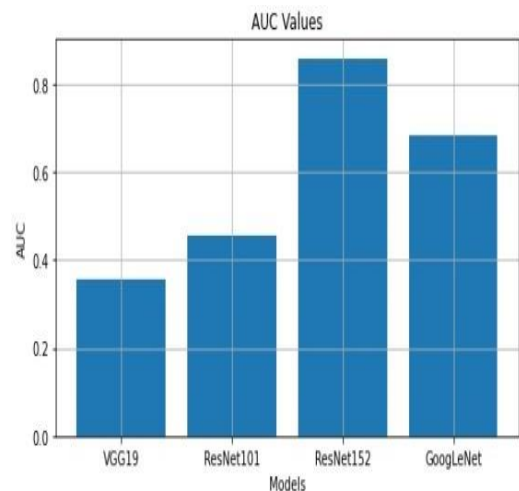


Fig. 11. AUC Values for each Model

Under the Curve (AUC) values for each model. This quantifiable measure provides a valuable assessment of their respective performance, with higher AUC values indicating enhanced classification accuracy.

After evaluating various performance metrics, including the confusion matrix, accuracy, ROC curve, and AUC values, it is evident that the ResNet152 model outperforms other approaches in oral cancer detection using the CNN project. The confusion matrix demonstrates that ResNet152 achieves a superior balance between true positives and true negatives, reducing both false positives and false negatives. Moreover, the accuracy of ResNet152 surpasses that of other models, indicating its ability to make correct predictions more consistently. The ROC curve analysis showcases a higher true positive rate and a lower false positive rate for ResNet152, signifying its excellent discriminatory power in distinguishing between cancerous and non-cancerous cases. Finally, the AUC value, being a comprehensive performance indicator, confirms that ResNet152 provides the best overall performance for oral cancer detection, making it the most promising and reliable choice for this critical medical application.

V. CONCLUSION AND FUTURE SCOPE

After evaluating multiple models, Resnet152 emerged as the top-performing model across all metrics, making it the preferred choice. Increasing at a rapid pace, the prevalence of Oral Potentially Malignant Disorders in India presents a concerning trend. Despite the numerous technological advancements achieved by researchers, oncologists, and dentists, the diagnostic process for oral cancer often occurs at an advanced stage, resulting in exorbitant expenses and disheartening outcomes. Essential progress in patient outcomes can only be achieved through the early detection and treatment of cancer, facilitated by innovative devices such as Vizilite, VELscope, and Cell Scope. Therefore, the timely identification and prevention of its progression to OSCC (Oral squamous cell carcinoma) is an urgent imperative. This paper aims to contribute to the future development of early detection methods using a standard phone camera, enabling the timely identification of malignant lesions in the oral cavity. To maximize the project's impact, future enhancements should prioritize expanding collaborative networks with healthcare professionals, research institutions, and oral health organizations. By fostering partnerships, collaborative efforts can tap into collective expertise, access larger datasets for training and validation, and gain valuable insights to refine the solution. This approach will lead to widespread adoption and continuous improvement of the system for early detection and prevention of oral cancer. Additionally, a key area for improvement lies in continuously refining the mobile application's user interface and user experience. Incorporating user feedback, conducting usability studies, and implementing iterative design processes can optimize the app's navigation, functionality, and overall usability. This will ensure a seamless and intuitive experience, encouraging greater engagement and adoption of the early detection features.

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